PYTHON PROJECT

Log File Analyzer

A simple tool to read, validate, and summarize log entries

Presented by: Drashy Raj

Class: IT-A (1B8)

Enroll No. : 12402080601040

Sem : 5th

Project Introduction

What the project does

- A Log File Analyzer is a Python program that reads a text log file.
- It checks whether each non-empty line follows the expected log format.
- It counts entries by log level: INFO, WARNING, ERROR, and DEBUG.
- It displays all ERROR messages separately for quick review.

Problem Statement

Why this tool is useful

- Log files can contain many lines, making manual checking time-consuming.
- Important error messages can be difficult to find in a large file.
- Incorrectly formatted lines may go unnoticed without validation.
- This project automates basic counting and error-message extraction.

Objectives

Main goals of the project

- Read log data from the file named sample.log.
- Recognize valid entries using a regular expression (Regex).
- Count valid entries and invalid entries.
- Summarize the number of entries for each log level.
- Print the messages that belong to the ERROR level.

Tools and Technologies

Simple Python concepts used

Python

Main programming language

re module

Checks each line with Regex

Text file

sample.log stores the log entries

The program uses standard Python features and does not require external libraries.

How the Program Works

Processing flow



If sample.log is missing, the program catches FileNotFoundError and prints a helpful message.

Log Format and Validation

What counts as a valid entry

Expected format

```
YYYY-MM-DD HH:MM:SS LEVEL Message
```

Example

```
2026-10-26 14:30:00 ERROR Unable to connect to server
```

- Allowed levels: INFO, WARNING, ERROR, DEBUG.
- Blank lines are skipped; lines that do not match are counted as invalid.

Important Code Components

Key parts of the Python script

1. Regular expression

```
pattern = r"^(date) (time) (level) (message)$"
```

2. Counting log levels

```
level = match.group(3)
log_count[level] += 1
```

3. Error handling

```
except FileNotFoundError:
print("sample.log was not found")
```

Regex is shown here in simplified form to make the presentation easier to understand.

Output and Results

What the program displays

```
===== LOG FILE ANALYZER =====
Total valid entries: <count>
  Invalid entries: <count>

    Log Level Summary:
      INFO : <count>
    WARNING : <count>
     ERROR : <count>
     DEBUG : <count>

     Error Messages:
    - <error message>

Analysis completed successfully.
```

- The exact counts depend on the contents of sample.log.
- The script prints each ERROR message it finds.
- If the file is missing, a clear error message is shown.

Thank You!

The Log File Analyzer demonstrates how Python can automate basic log checking, counting, and error extraction.